Question no.1

There are three juniors and three seniors in the Service Club. The club is to send four representatives to the State Conference. How many different ways are there to select a group of four students to attend the conference?

Solution:

Choose 4 students from the total number of students. Order is not important. There are 6 total students. So number of choices is ${}^{6}C_{4} = 15$

Question no.2

There are four juniors and three seniors in the Service Club.The club is to send three representatives to the State Conference. If the members of the club decide to send one junior and two seniors, how many different groupings are possible?

Solution:

Choose 1 junior and 2 seniors: $4C_1 \cdot 3C_2 = 12$

Question no.3

Suppose that X = {*a*, *b*,*c*}. Then list all 3-combinations of the set *X*, and hence find the value of *C*(3,3).

Solution:

X = {*a*, *b*,*c*}.

Then 3-combinations of the 3 elements of the set *X* is: {*a*, *b*,*c*}.

Hence *C*(3,3) = 1.

Question no.4

If *W* = {*x*, *y*, *z*, *t*}. Then list all 3-combinations of the set *W*, and hence find the value of *C*(4,3).

Solution:

W = {*x*, *y*, *z*,*t*}.

Then 3-combinations of the 4 elements of the set *X* are:

{*x*,*y*,*z*}, {*x*, *y*,*t*},{x,z,t} and {*y*, *z*,*t*}.

Hence *C*(4,3) = 4.

Question no.5

Compute: $C(9, 6)$.

Solution:

$$\because \quad {}^{n}C_{r} = \frac{n!}{(n-r)!\,r!}$$

$$\Rightarrow \quad {}^{9}C_{6} = \frac{9!}{(9-6)!\,6!}$$

$$= \frac{9!}{3!\,6!}$$

$$= \frac{362880}{6.(720)}$$

$$= \frac{362880}{4320}$$

$$= 84$$

Question no.6

There are total 14 players in a cricket team. The captain needs to choose 10 players to play in a match. How many possible teams can be chosen?

Solution:

This is a combination problem as the order in which the teams are chosen is not important. The number of combinations is

$$\because \quad {}^{n}C_{r} = \frac{n!}{(n-r)!\,r!}$$

$$\Rightarrow \quad {}^{14}C_{10} = \frac{14!}{(14-10)!\,10!}$$

$$= \frac{14!}{4!\,10!}$$

$$= \frac{87178291200}{24.2628800}$$

$$= 1001$$

Question no.7

In how many ways can a team of 5 candidate can be formed out of a total of 12 candidates such that two particular candidates should be included in each department?

Solution:

Two particular candidates should be included in each team.

i.e., we have to select 5-2 = 3 persons from 12-2 = 10persons

Hence, the required number of ways are:

$$\because \quad {}^{n}C_{r} = \frac{n!}{(n-r)!\,r!}$$

$$\Rightarrow \quad {}^{10}C_{3} = \frac{10!}{(10-3)!\,3!}$$

$$= \frac{10!}{7!\,3!} = 120$$